

भारतीय मानक
Indian Standard

IS 302 (Part 2/Sec 7) : 2024

घरेलू और समान विद्युत साधित्रों की सुरक्षा
भाग 2 विशेष अपेक्षाएँ
अनुभाग 7 कपड़े धोने की मशीन
(दूसरा पुनरीक्षण)

**Safety of Household and Similar
Electrical Appliances**
Part 2 Particular Requirements
**Section 7 Domestic Electric Clothes Washing
Machines**
(Second Revision)

ICS 97.060; 13.120

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FOREWORD

This Indian Standard (Part 2/Sec 7) (Second Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council.

This standard was first published in 1994 and revised in 2010. This revision has been undertaken primarily to align the existing standard with the latest International Standard.

This standard covers the general and safety requirements of domestic electric clothes washing machines. This standard however does not cover the performance requirements, which is covered under a separate composite standard, IS 14155 'Domestic electric clothes washing machines for household use'.

This standard is to be read in conjunction with the latest edition of IS 302 (Part 1) 'Safety of household and similar electrical appliances: Part 1 General Requirements' and its amendments. This standard was formulated on the basis of IS 302-1 : 2008.

NOTE — When "Part 1" is mentioned in this standard, it refers to IS 302-1.

This Part 2 supplements or modifies the corresponding clauses in IS 302 (Part 1), so as to convert that standard into the Indian standard: Particular requirements for domestic electric clothes washing machines. When a particular subclause of Part 1 is not mentioned in this Part 2, that subclause applies as far as is reasonable. When this standard states addition, modification or replacement, the relevant text in Part 1 is to be adapted accordingly.

NOTE — The following numbering system is used:

- a) Sub clauses, tables and figures that are numbered starting from 101 are additional to those in Part 1; and
- b) Additional annexes are lettered AA, BB, etc.

This standard is based on IEC 60335-2-7: 2019 (Ed. 8.0). As this standard refers to IS 302 (Part 1), the differences of IS 302 (Part 1) from IEC 60335-1 shall apply.

The principal changes in this revision are as follows (minor changes are not listed):

- a) requirements for steam generators in **3.1.9, 3.6.101, 19.101, 22.106, 22.107**;
- b) additional definitions for washing machine types in **3.5.101, 3.5.102, 3.5.103**;
- c) revised temperature limits for external accessible surfaces in **11.3** and **11.8**;
- d) revised test procedure for the spillage test in **15.2**; and
- e) additional requirements for appliances that are controlled by programmable electronic circuits that limit the number of heating elements and motors from being energised at the same time, Sub clause **22.108**.

It has been assumed in the formulation of this standard that the execution of its provisions is entrusted to appropriately qualified and experienced persons.

This standard recognizes the internationally accepted level of protection against hazards such as electrical, mechanical, thermal, fire and radiation of appliances when operated as in normal use taking into account the manufacturer's instructions. It also covers abnormal situations that can be expected in practice and takes into account the way in which electromagnetic phenomena can affect the safe operation of appliances.

This standard takes into account the requirements of IS 732 : 2019 'Code of practice for electrical wiring installations (*fourth revision*)', as far as possible so that there is compatibility with the wiring rules when the appliance is connected to the supply mains. However, in case of any deviation, wiring rules take precedence.

(Continued on third cover)

Indian Standard
**HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES –
SAFETY**

PART 2 PARTICULAR REQUIREMENTS

SECTION 7 DOMESTIC ELECTRIC CLOTHES WASHING MACHINES

(*Second Revision*)

1 SCOPE

This clause of Part 1 is replaced by the following.

This standard deals with the safety of electric washing machines for household and similar use, that are intended for washing clothes and textiles, their rated voltage being not more than 250 V for single-phase appliances and 480 V for other appliances.

This standard also deals with the safety of electric washing machines for household and similar use employing an electrolyte instead of detergent. Additional requirements for these appliances are given in Annex CC.

NOTE — Guidance is given in Annex DD for requirements that can be used to ensure an acceptable level of protection against electrical and thermal hazards for washing machines fitted with a power driven wringer.

Appliances not intended for normal household use but which nevertheless may be a source of danger to the public, such as appliances intended to be used by laymen in shops, in light industry and on farms, are within the scope of this standard.

NOTE — Examples of such appliances are washing machines for communal use in blocks of flats or in launderettes.

As far as is practicable, this standard deals with the common hazards presented by washing machines that are encountered by all persons in and around the home. However, in general, it does not take into account:

- a) persons (including children) whose:
 - 1) physical, sensory or mental capabilities; or
 - 2) lack of experience and knowledge prevents them from using the appliance safely without supervision or instruction; and
- b) children playing with the appliance.

NOTES

- 1 Attention is drawn to the fact that:
 - a) for washing machines intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary; and
 - b) in many countries additional requirements are specified by the national health authorities, the national authorities responsible for the protection of labour, the national water supply authorities and similar authorities.
- 2 This standard does not apply to:
 - a) washing machines intended exclusively for industrial purposes (ISO 10472-2);
 - b) appliances intended to be used in locations where special conditions prevail, such as the presence of a corrosive or explosive atmosphere (dust, vapour or gas); and
 - c) washing machines incorporating steam generating devices in which steam is produced at a pressure exceeding 50 kPa.

2 NORMATIVE REFERENCES

This clause of Part 1 is applicable except as follows.

Addition:

<i>IS No.</i>	<i>Title</i>
IS 16923 (Part 1) : 2018/ IEC 60584-1 : 2013	Thermocouples: Part 1 EMF specifications and tolerances (<i>first revision</i>)
IS 3400 (Part 6) : 2018/ ISO 1817 : 2015	Methods of test for vulcanized rubbers: Part 6 Determination of the effect of liquids (<i>fourth revision</i>)
IEC 60730-2-12 : 2015	Automatic electrical controls — Part 2-12: Particular requirements for electrically operated door locks

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3 TERMS AND DEFINITIONS

This clause of Part 1 is applicable except as follows.

3.1.9 Normal Operation — operation of the appliance under the following conditions.

The appliance is filled with dry textile material having a mass equal to the maximum mass stated in the instructions, and with the maximum quantity of water for which it is constructed. However, if the power input or current is higher when only 50 percent of the textile material is used, the appliance is operated with this load instead if this gives more unfavourable conditions than the full load during the test of **11**.

NOTE — For some appliances incorporating a programmer, using the 50 percent reduced load may result in automatic selection of a reduced wash programme.

The temperature of the water is:

- a) $(65 \pm 5)^\circ\text{C}$ for appliances without heating elements;
- b) $(15 \pm 5)^\circ\text{C}$ for appliances without heating elements and intended for connection to the cold water supply only; and
- c) $(15 \pm 5)^\circ\text{C}$ for other appliances.

If the appliance does not incorporate a programmer, the water is heated to $(90 \pm 5)^\circ\text{C}$ or as high as the construction will allow if lower, before starting the first washing period.

The textile material consists of pre-washed double-hemmed cotton sheets having dimensions approximately $700\text{ mm} \times 700\text{ mm}$ and a specific mass between 140 g/m^2 and 175 g/m^2 in the dry condition.

For impeller washing machines, if the textile material does not move properly during operation:

- a) the quantity of textile material may be reduced until the maximum power input of the motor is attained; or
- b) a textile material comprising pre-washed double-hemmed cotton sheets, having dimensions of approximately $900\text{ mm} \times 900\text{ mm}$ and a mass between 90 g/m^2 and 110 g/m^2 in the dry condition, may be used.

However, for impeller washing machines, in case of doubt, the test is carried out using the reduced quantity of textile material.

A steam generator intended to be filled by hand is filled according to the instructions, water being added to maintain the steam generation.

A steam generator intended to be filled automatically is connected to the water mains.

Additional Sub-clauses

3.5.101 Agitator Washing Machine — Washing machine in which the textiles are substantially immersed in the washing water, the mechanical action being produced by a device moving about or along its vertical axis with a reciprocating motion (an agitator).

NOTE — This device usually extends above the maximum water level.

3.5.102 Impeller Washing Machine — Washing machine in which the textiles are substantially immersed in the washing water, the mechanical action being produced by a device rotating about its axis continuously or which reverses after a number of revolutions (an impeller)

NOTE — The uppermost point of this device is substantially below the minimum water level.

3.5.103 Drum Washing Machine — Washing machine in which the textiles are placed in either a horizontal drum or a drum that is inclined up to and including 45° from the horizontal and the textile is partially immersed in the washing water, the mechanical action being produced by rotation of the drum about its axis, the movement being either continuous or periodically reversed

3.6.101 Steam Generator — Device in which steam is produced at a pressure not exceeding 50 kPa and in which the pressure drops to atmospheric pressure when the steam is not supplied.

4 GENERAL REQUIREMENT

This clause of Part 1 is applicable.

5 GENERAL CONDITIONS FOR THE TESTS

This clause of Part 1 is applicable except as follows.

5.2 Addition

The relevant tests of **21.101**, **21.102** and **22.104** shall be carried out on the same appliance as that used for the test of **18**.

5.3 Addition

The test of **15.101** is carried out before the test of **15.3**.

The relevant tests of **21.101** and **21.102** are carried out before the test of **18**. The test of **22.104** is carried out after the test of **18**.

5.7 Addition

Doubt is considered to exist if the temperature of the water is within 6 K of the boiling point and the difference between the temperature rise of the

relevant part and the limit specified does not exceed 25 K minus the room temperature.

6 CLASSIFICATION

This clause of Part 1 is applicable except as follows.

6.1 Modification

Appliances shall be of Class I, Class II or Class III.

6.2 Addition

Appliances shall be at least IPX4.

7 MARKING AND INSTRUCTIONS

This clause of Part 1 is applicable except as follows.

7.1 Addition

Appliances without automatic water level control shall be marked with the maximum water level.

Appliances not intended for connection to the hot water supply and not provided with heating elements shall be marked with the substance of the following:

CAUTION — Do not connect to the hot water supply.

7.10 Addition

If the off position is only indicated by letters, the word "off" shall be used.

7.12 Addition

The instructions shall specify the maximum mass of dry cloth in kilograms to be used in the appliance.

The instructions shall include the substance of the following:

This appliance is intended to be used in household and similar applications such as:

- a) staff kitchen areas in shops, offices and other working environments;
- b) farm houses;
- c) by clients in hotels, motels and other residential type environments;
- d) bed and breakfast type environments; and
- e) areas for communal use in blocks of flats or in launderettes.

If the manufacturer wants to limit the use of the appliance to less than the above, this shall be clearly stated in the instructions.

7.12.1 Addition

For washing machines having ventilation openings in the base, the installation instructions shall state that the openings must not be obstructed by a carpet.

7.15 Addition

The caution relating to connection to the hot water supply shall be on the appliance at its point of attachment to the water supply.

8 PROTECTION AGAINST ACCESS TO LIVE PARTS

This clause of Part 1 is applicable.

9 STARTING OF MOTOR-OPERATED APPLIANCES

This clause of Part 1 is not applicable.

10 POWER INPUT AND CURRENT

This clause of Part 1 is applicable except as follows.

10.1 Addition

The selected representative period is the period, such as filling with water, washing, rinsing, water extraction, spinning or braking, during which the power input is the highest.

10.2 Addition

The selected representative period is the period, such as filling with water, washing, rinsing, water extraction, spinning or braking, during which the current is the highest.

11 HEATING

This clause of Part 1 is applicable except as follows.

11.3 Addition

Where the external accessible surfaces are suitably flat and access permits, then the test probe of Figure 101 may be used to measure the temperature rises of external accessible surfaces specified in Table 101. The probe is applied with a force of (4 ± 1) N to the surface in such a way that the best possible contact between the probe and the surface is ensured. The measurement is performed after a contact period of 30 s.

The probe may be held in place using a laboratory stand clamp or similar device. Any measuring instrument giving the same results as the probe may be used.

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11.7 Replacement

Appliances incorporating a programmer are operated for three cycles with the programme that results in highest temperature rises, with a rest period of 4 min between cycles.

Other appliances are operated for three cycles, with a rest period of 4 min between cycles. Each cycle consists of the following operations:

a) for appliances without means for water extraction and for washing machines with a hand-operated wringer,	washing;
b) for appliances having a single drum for washing and water extraction,	washing followed by water extraction;
c) for appliances having separate drums for washing and water extraction that cannot be used simultaneously,	washing and water extraction separated by an additional 4 min rest period;
d) for appliances having separate drums for washing and water extraction that can be used simultaneously,	washing together with water extraction so that the operations terminate simultaneously;
e) for appliances having a single drum for washing, water extraction and drying 1) that allow the same quantity of textile material to be washed and dried in the drum, 2) that, according to the instructions, only allow a portion of the washed textile material to be dried in the drum,	washing followed by water extraction, followed by drying; washing followed by water extraction followed by two drying periods, with an additional rest period of 4 min before each drying period. In this case only two cycles of operation are carried out.

For appliances incorporating a timer, the washing period, the water extraction period and the drying period are equal to the maximum period allowed by the timer.

For appliances without a timer,

- a) the washing period has a duration of:
 - 1) 6 min, for impeller washing machines;
 - 2) 18 min, for agitator washing machines; and
 - 3) 25 min for drum washing machines unless a longer period is stated in the instructions;
- b) the water extraction period has a duration of 5 min.

The rest period, including any braking time, has a duration of 4 min.

After the specified sequence of operation, discharge pumps that are driven by a separate motor and switched on and off manually, are subjected to three operating periods separated by rest periods of 4 min. Each operating period is equal to 1.5 times the period necessary to empty the appliance when filled to the maximum normal water level. The outlet of the water discharge pipe is 900 mm above the floor.

11.8 Addition

During the test, the temperature rises are monitored continuously for one cycle and shall not exceed the values shown in Table 101.

Table 101 Maximum Temperature Rises for External Accessible Surfaces under Normal Operating Conditions

(Clause 11.8)

SI No.	Surface	Temperature rise of External Accessible Surfaces ^a		
		K		
		Surfaces of appliances situated not more than 850 mm above the floor after installation		Surfaces situated more than 850 mm above the floor after installation ^b
		Front surfaces	Other surfaces ^b	
(1)	(2)	(3)	(4)	(5)
i)	Bare metal	38	42	42
ii)	Coated metal ^c	42	49	49
iii)	Glass and ceramic	51	56	56
iv)	Plastic and plastic coating > 0.4 mm ^{d, e}	58	62	62

^a Temperature rises are not measured on

- the underside of appliances intended to be used on a working surface or floor, where these surfaces are inaccessible to a 75 mm diameter probe having a hemispherical end, applied with a force not exceeding 1 N,
- the rear surface of appliances which, according to the instructions, shall be placed against a wall and where these surfaces are inaccessible to a 75 mm diameter probe having a hemispherical end, applied with a force not exceeding 1 N,
- hot water supply fittings and hoses,
- surfaces within 25 mm from air outlets valves can be increased by 10 K.

^b If these values are exceeded, the test is allowed to be repeated with the appliance moved away from the test corner wall. The test is repeated for 1 cycle.

^c Metal is considered coated when a coating having a minimum thickness of 90 µm made by enamel or non-substantially plastic coating is used.

^d The temperature rise limit of plastic also applies for plastic material having a metal finish of thickness less than 0.1 mm.

^e When the thickness of the plastic coating does not exceed 0.4 mm, the temperature rise limits of coated metal for underlying metal apply or the temperature rise limits for glass or ceramic material for underlying glass or ceramic material apply.

12 VOID

13 LEAKAGE CURRENT AND ELECTRIC STRENGTH AT OPERATING TEMPERATURE

This clause of Part 1 is applicable except as follows.

13.2 Modification

For stationary class I appliances, the leakage current shall not exceed 3.5 mA or 1 mA/kW of rated power input with a limit of 5 mA, whichever is greater.

14 TRANSIENT OVERVOLTAGES

This clause of Part 1 is applicable.

15 MOISTURE RESISTANCE

This clause of Part 1 is applicable except as follows.

15.2 Replacement

Appliances shall be constructed so that spillage of liquid in normal use does not affect their electrical insulation even if an inlet valve fails to close.

Compliance is checked by the following test.

Appliances with type X attachment, except those having a specially prepared cord, are fitted with the lightest permissible type of flexible cord of the smallest cross-sectional area specified in Table 13.

Appliances intended to be filled with water by the user are completely filled with water containing approximately 1 percent NaCl. A further quantity of this solution equal to 15 percent of the capacity of the appliance or 0.25 liter, whichever is greater, is poured in steadily over a period of 1 min.

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Other appliances are operated until the maximum water level is reached, and 5 g of the detergent specified in Annex AA is added for each litre of water in the appliance. Each inlet valve is held open and the filling is continued for 15 min after first evidence of overflow or until the inflow is automatically stopped by other means. The appropriate tests being applied one at a time to each inlet valve.

For appliances that are loaded from the front, the door is then opened if this can be achieved manually and without damage to the door interlock system.

For all appliances, 0.5 litre of a solution comprising water containing approximately 1 percent NaCl and 0.6 percent of rinsing agent, is rapidly poured over the top of the appliance so that the spillage solution flows over the surfaces of the appliance that incorporate controls, the controls being placed in the on position. The controls are then operated through their working range, this operation being repeated after a period of 5 min.

Any commercially available non-ionic rinsing agent may be used, but if there is any doubt with regards to the test results, the rinsing agent shall have the following properties:

- a) Viscosity — 17 mPa·s
- b) pH — 2.2 (1 percent in water)

and its composition shall be as given below

<i>Sl No.</i>	<i>Substance</i>	<i>Parts by mass percent</i>
(1)	(2)	(3)
i)	Plurafac ® LF 221	15.0
ii)	Cumene sulfonate (40 percent solution)	11.5
iii)	Citric acid (anhydrous)	3.0
iv)	Deionized water	70.5

NOTE — Plurafac ® LF 221 is the trade name of a product supplied by BASF. This information is given for the convenience of users of the standard and does not constitute an endorsement.

The appliance shall then withstand the electric strength test of **16.3** and inspection shall show that there is no trace of water on insulation that could result in a reduction of clearances or creepage distances below the values specified in **29**.

15.101 Appliances shall be constructed so that foaming does not affect electrical insulation.

Compliance is checked by the following test that is carried out immediately after that of percent.

The appliance is operated under the conditions specified in **11** but at rated voltage for one complete cycle with the programme that results in the longest period of operation. A quantity of detergent necessary to cause foaming is added. The composition of the detergent is specified in Annex AA.

For appliances incorporating a detergent dispenser, the solution is added manually at the point in the cycle when it would normally be dispensed automatically. For other appliances, the solution is added before starting the cycle.

The appliance shall then withstand the electric strength test of **16.3**.

The appliance is kept in a test room having a normal atmosphere for 24 h before being subjected to the test of **15.3**.

16 LEAKAGE CURRENT AND ELECTRIC STRENGTH

This clause of Part 1 is applicable.

17 OVERLOAD PROTECTION OF TRANSFORMERS AND ASSOCIATED CIRCUITS

This clause of Part 1 is applicable.

18 ENDURANCE

This clause of Part 1 is replaced by the following.

18.101 Appliances shall be constructed so that the lid or door interlock withstands the stresses to which it may be exposed in normal use.

Compliance is checked by the following test.

The lid or door is opened as in normal use and the force applied to the handle, or actuating means of the release mechanism, is measured. The force required to close the lid or door is also measured.

The lid or door is then subjected to 10 000 cycles of opening and closing. For the first 6 000 cycles, the appliance is supplied at rated voltage and operated so that the interlock mechanism is energized and de-energized each cycle. For the last 4 000 cycles, the appliance is not connected to the supply mains. For appliances having a drying function, the total number of cycles is increased to 13 000, the first 9 000 cycles being carried out with the interlock mechanism energized and de-energized each cycle.

If the interlock complies with IEC 60730-2-12, the appliance is not connected to the supply mains during this test. If the interlock operates more than once during normal operation, it is operated for this number of times during each cycle.

Lids are opened each time by approximately 45° and doors by 90°, the speed of opening being approximately 1.5 m/s. The force applied to open the lid or door is twice the measured opening force, with a minimum of 50 N and a maximum of 200 N.

Doors are closed at a speed of approximately 1.5 m/s, the force applied being five times the measured closing force, with a minimum of 50 N and a maximum of 200 N. Lids are allowed to close under their own weight but if they fail to latch, a force of five times the measured closing force is applied, with a minimum of 50 N and a maximum of 200 N.

After the tests, compliance with the relevant requirements of **20.103** to **20.105** shall not be impaired.

18.102 The braking mechanism of appliances having a lid that can be opened during the water extraction period shall withstand the stresses to which they may be exposed in normal use.

Compliance is checked by the following test.

The appliance is supplied at 1.06 times rated voltage and operated under normal operation until the motor has reached its highest speed. The lid is then fully opened. The test is repeated after the drum has been at rest for a period long enough to ensure that the appliance does not attain an excessive temperature.

The test is carried out 1 000 times, the textile material being re-saturated with water at least every 250 times.

After the test, the appliance shall be fit for further use and compliance with this standard shall not be impaired.

NOTE — Forced cooling can be used to prevent excessive temperatures and to shorten the test.

19 ABNORMAL OPERATION

This clause of Part 1 is applicable except as follows.

19.1 Addition:

For appliances incorporating a programmer or a timer, the tests of **19.2** and **19.3** are replaced by the test of **19.101**.

The test of **19.7** is not carried out on motors driving moving parts of an oscillating agitator.

Appliances not intended for connection to the hot water supply and not provided with heating elements are also subjected to the test of **19.102**.

19.2 Addition:

Restricted heat dissipation is obtained without water in the appliance or with just sufficient water to cover the heating elements, whichever is the more unfavourable.

19.7 Addition:

Appliances without a programmer or timer are operated for 5 min.

19.9 Addition:

The running overload test is carried out on appliances that have overload protective devices incorporating electronic circuits to protect the windings of the drum motor. However, the test is not carried out if the protective device senses the winding temperature directly.

NOTE — Measuring winding resistance or winding current is not directly measuring the winding temperature.

The appliance is operated under the conditions of **11** for one cycle. The load is then increased so that the current through the motor windings is raised by 10 percent. The appliance is operated again for the same cycle; the supply voltage being maintained at its original value. The load is again increased and the test is repeated until the protective device incorporating the electronic circuit operates or the motor stalls.

19.13 Addition:

The textile material shall not ignite and shall not show any charring or glowing.

NOTE — Light brown colouring of the textile material or slight emission of smoke can be ignored.

During the tests of **19.101** and **19.102**, the temperature of windings shall not exceed the values specified in Table 8.

The appliance shall comply with the appropriate requirements of **20.103** to **20.105** if it can still be operated.

19.101 The appliance is supplied at rated voltage and operated under normal operation. Any fault condition or unexpected operation that may be applied in normal use is introduced.

The fault conditions and unexpected operations to be applied are:

- a) the programmer stopping in any position;

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- b) disconnection and reconnection of one or more phases of the supply during any part of the programme;
- c) open-circuiting or short-circuiting of components;
- d) failure of a magnetic valve;
- e) failure or blocking the mechanical parts of a water-level switch. This fault condition is not applied if:
 - 1) the cross-sectional area of the tube supplying the air chamber is greater than 500 mm² with a minimum dimension of 10 mm;
 - 2) the outlet of the chamber is at least 20 mm above the highest water level; and
 - 3) the tube connecting the air chamber to the water-level switch is fixed so that there is no likelihood of bending or pinching;
- f) puncture of the capillary tube of a thermostat; and
- g) the steam generator is operating without water.

If operation without water in the appliance is a more unfavourable condition for starting any programme, the tests with that programme are carried out with the water valve closed. This valve is not closed after the programme has started to operate.

NOTES — The fault condition with:

- 1 the automatic filling device held open is covered by 15.2;
- 2 thermal controls short-circuited is covered by 19.4;
- 3 motor capacitors short-circuited or open-circuited is covered by 19.7; and
- 4 the failure of door interlocks is covered by 24.1.4.

19.102 Appliances not intended for connection to the hot water supply and not provided with heating elements are operated under the conditions of 11, except that they are supplied at rated voltage and filled with water at a temperature of $(65 \pm 5) ^\circ\text{C}$.

20 STABILITY AND MECHANICAL HAZARDS

This clause of Part 1 is applicable except as follows.

20.1 Modification

The appliance is empty or filled as specified for normal operation, whichever is more unfavourable. Doors and lids are closed and any castors turned to the most unfavourable position.

20.101 Drum washing machines that are loaded from the top through an opening with a hinged lid shall incorporate an interlock that de-energizes the motor before the lid opening exceeds 50 mm.

If a removable or sliding lid is provided, the motor shall be de-energized as soon as the lid is removed or displaced and it shall not be possible to start the motor unless the lid is in the closed position.

The interlock shall be constructed so that unexpected operation of the appliance is unlikely unless the lid is in the closed position.

Compliance is checked by inspection, by measurement and by the following test.

Test probe B of IEC 61032 is applied in order to try and release any interlock that is needed to comply with the requirement. The interlock shall not release.

20.102 Appliances shall not be adversely affected by an unbalanced load.

Compliance is checked by the following test.

The appliance is placed on a horizontal support and a load having a mass of 0.2 kg or 10 percent of the maximum mass of the cloth specified in the instructions, whichever is greater, is fixed to the inside wall of the drum half-way along its length.

The appliance is supplied at rated voltage and operated during the water extraction period.

The test is carried out four times, the load being moved each time through an angle of 90° around the wall of the drum.

If compliance relies on the operation of an electronic circuit, the test is repeated with the fault conditions in a) to g) of 19.11.2 applied one at a time to the electronic circuit.

The appliance shall not overturn and the drum shall not hit other parts except the enclosure. After the test, the appliance shall be fit for further use.

20.103 Drum washing machines that are loaded from the front or from the top, the door or lid shall be interlocked so that the appliance can only be operated when the door or lid is in the closed position.

Compliance is checked by inspection, by manual test and by the following test.

Test probe B of IEC 61032 is applied in order to try and release any interlock that is needed to comply with the requirement. The interlock shall not release.

20.104 It shall not be possible to open the lid or door of the appliance while the drum speed exceeds 60 r/min if the drum has a rotational kinetic energy exceeding 1 500 J, or a maximum peripheral speed exceeding:

- a) 20 m/s, for drums that rotate about the horizontal axis or an axis inclined up to and

including 45° from the horizontal; and

- b) 40 m/s, for drums that rotate about the vertical axis.

Compliance is checked by the following test.

The appliance is supplied at rated voltage and operated empty. The force determined during the test of **22.104** with the lid interlocked is applied to the lid or door in an attempt to open it.

If compliance relies on the operation of an electronic circuit, the test is repeated under the following conditions applied separately:

- a) the fault conditions in a) to g) of **19.11.2** applied one at a time to the electronic circuit; and
- b) the electromagnetic phenomena tests of **19.11.4.2** and **19.11.4.5** applied to the appliance.

It shall not be possible to open the lid or door while the drum speed exceeds 60 r/min. If the appliance is loaded from the front and the door can be opened, the motor shall be de-energized before the opening exceeds 50 mm.

NOTE — The rotational kinetic energy can be calculated from the following formula:

$$E = mv^2/4$$

where

- E = rotational kinetic energy, in J;
- m = mass of cloth specified in the instructions, in kg; and
- v = maximum peripheral speed of the drum, in m/s.

If the electronic circuit is programmable, the software shall contain measures to control the fault/error conditions specified in Table R.1 and is evaluated in accordance with the relevant requirements of Annex R.

20.105 Appliances shall have an automatic means for switching off the motor, or for reducing the drum speed to 60 r/min, when the lid or door is opened if the drum has a rotational kinetic energy not exceeding 1 500 J and a peripheral speed not exceeding

- a) 20 m/s, for drums that rotate about the horizontal axis or an axis inclined up to and including 45° from the horizontal; and
- b) 40 m/s, for drums that rotate about the vertical axis.

NOTE — The rotational kinetic energy is calculated in accordance with the formula in **20.104**.

Compliance is checked by the following test.

The appliance is supplied at rated voltage and operated in accordance to **20.1** empty or filled as specified for normal operation, whichever is more unfavourable. A force not exceeding 50 N is applied to the lid or door in an attempt to open it, as in normal use.

If compliance relies on the operation of an electronic circuit, the test is repeated under the following conditions applied separately:

- a) the fault conditions in a) to g) of **19.11.2** applied one at a time to the electronic circuit; and
- b) the electromagnetic phenomena tests of **19.11.4.2** and **19.11.4.5** applied to the appliance.

If the lid or door opens, the drum speed shall be no higher than 60 r/min within 7 s of opening the lid or door by 50 mm. In addition, if the appliance is loaded from the front, the motor shall become de-energized.

If the electronic circuit is programmable, the software shall contain measures to control the fault/error conditions specified in Table R.1 and is evaluated in accordance with the relevant requirements of Annex R.

20.106 For appliances with a front opening door having an opening dimension exceeding 200 mm, and drum volume exceeding 60 dm³, it shall not be possible to start or recommence the washing cycle until a separate means which controls the movement of the drum is operated manually, even after the door has been opened and closed again.

NOTE — The volume of the drum can be calculated by measuring the maximum internal diameter and maximum internal length of the drum.

Compliance is checked by inspection, measurement ignoring any non-metallic seal fitted in the door opening, and by the following test.

The appliance is supplied at rated voltage, and the door is opened and then closed.

If compliance relies on the operation of an electronic circuit, the test is repeated under the following conditions applied separately:

- a) the fault conditions in a) to g) of **19.11.2** are applied one at a time to the electronic circuit; and
- b) the electromagnetic phenomena tests of **19.11.4.2** and **19.11.4.5** are applied to the appliance.

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The washing cycle shall not start or recommence.

20.107 For appliances with a front opening door having an opening dimension exceeding 200 mm, and drum volume exceeding 60 dm³, it shall be possible to open from the inside the closed door, when the appliance is not energized or in a standby mode, with a force not exceeding 70 N.

NOTE — The volume of the drum can be calculated by measuring the maximum internal diameter and maximum internal length of the drum.

Compliance is checked by measurement, ignoring any non-metallic seal fitted in the door opening, and by applying a force of 70 N perpendicular to the plane of the closed door at a point furthest from the hinges accessible from the inside of the door. If the appliance is supplied with an additional decorative door, the test is carried out with this door closed.

NOTE — The force can be applied to the outside of the door.

21 MECHANICAL STRENGTH

This clause of Part 1 is applicable except as follows.

21.101 Lids and doors shall have adequate mechanical strength.

Compliance is checked by the test of **21.101.1** for lids and **21.101.2** for doors.

21.101.1 A rubber hemisphere having a diameter of 70 mm and a hardness between 40 IRHD and 50 IRHD is fixed to a cylinder having a mass of 20 kg and dropped from a height of 100 mm onto the centre of the lid.

The test is carried out three times, after which the lid shall not be damaged to such an extent that moving parts become accessible.

21.101.2 A vertically downwards force of 150 N is applied in the most unfavourable position to the door while it is open at an angle of $90^\circ \pm 5^\circ$. The force is maintained for 1 min.

After the test, the appliance shall not be damaged or deformed to such an extent that compliance with **20.103** to **20.105** is impaired.

21.102 Lids shall have adequate resistance to distortion.

Compliance is checked by the following test.

A force of 50 N is applied to the open lid in the most unfavourable direction and position.

The test is carried out three times, after which the hinges shall not have worked loose and the

appliance shall not be damaged or deformed to such an extent that compliance with **20.103** to **20.105** is impaired.

22 CONSTRUCTION

This clause of Part 1 is applicable except as follows.

22.6 Modification to the Requirement:

The requirement relating to leakage from containers, hoses, couplings and similar parts of the appliance is not applicable to parts that withstand the ageing test specified in Annex BB.

Modification to the test specification:

Instead of coloured water, a solution composed of 5 g of the detergent specified in Annex AA per litre of distilled water is used.

22.101 Appliances shall be constructed so that when the water level is above the lower edge of the door opening, it shall not be possible to open the door by a simple action while the appliance is operating. This requirement is not applicable to appliances fitted with interlocked doors or doors that are opened by means of a key or by two separate actions, such as pushing and turning.

Compliance is checked by inspection and by manual test.

If compliance relies on the operation of an electronic circuit and the appliance is capable of providing a wash water temperature of 60 °C or higher or is marked as having a wash water temperature of 60 °C or higher, the test is repeated under the following conditions applied separately:

- a) the fault conditions in a) to g) of **19.11.2** applied one at a time to the electronic circuit; and
- b) the electromagnetic phenomena tests of **19.11.4.2** and **19.11.4.5** applied to the appliance.

It shall not be possible to open the lid or door by a simple action.

If the electronic circuit is programmable, the software shall contain measures to control the fault/error conditions specified in Table R.1 and is evaluated in accordance with the relevant requirements of Annex R.

22.102 Appliances shall be constructed so that textile material cannot come into contact with heating elements.

Compliance is checked by inspection.

22.103 Appliances shall be constructed so that, during normal use, filter compartments cannot be opened by a simple action. This requirement is not applicable to appliances intended for connection to the cold water supply only and without means to heat the water or to appliances fitted with filter compartment covers that are:

- a) interlocked;
- b) opened by means of a key;
- c) opened by two separate actions such as pushing and turning; or
- d) opened by rotating by more than 180°.

Compliance is checked by inspection and by manual test.

22.104 Lid and door interlocks required for compliance with **20** shall be constructed so that they are unlikely to be forced open in normal use.

Compliance is checked by the following test.

The lid or door is opened as in normal use and the force applied to the handle, or actuating means of the release mechanism, is measured.

The lid and door is closed. The appliance is supplied at rated voltage and operated for a sufficient period for the interlock to be energized. An attempt is then made to open the lid or door as in normal use. The force applied is gradually increased to five times the measured opening force, with a minimum of 50 N and a maximum of 200 N, over a period of 5 s.

The test is carried out 300 times at a rate of approximately six times per minute.

The force is then increased to 10 times the measured opening force, with a minimum of 50 N. It shall not be possible to open the lid or door.

NOTE — Damage to handles is ignored.

22.105 Any mechanical release mechanism intended to open the loading door after a failure shall only be accessible by using a tool.

Compliance is checked by inspection.

22.106 Steam generators shall be vented to the atmosphere. The aperture shall be at least 5 mm in diameter or at least 20 mm² in area with a minimum dimension of 3 mm.

Compliance is checked by inspection and by measurement.

22.107 Appliances with steam generators shall be constructed in such a way that there is no spillage of water or sudden jets of steam or hot water likely to

expose the user to a hazard when the appliance is used in accordance with the instructions.

If jets of steam or liquids are emitted through protective devices, the electrical insulation shall not be affected or the user exposed to a hazard.

Compliance is checked by inspection and by inspection during the tests of **11** and **19**.

22.108 For appliances that are controlled by programmable electronic circuits that limit the number of heating elements and motors from being energised at the same time, simultaneous activation of any combination of heating elements and motors shall not render the appliance unsafe.

Compliance is checked as follows:

- a) the fault/error conditions specified in Table R.1 are applied and evaluated in accordance with the relevant requirements of Annex R; or
- b) the appliance is operated under the conditions of **11** while being supplied at rated voltage, the programmable electronic circuits being modified to allow simultaneous activation of all heaters and motors under their control. Under these conditions, compliance with **19.13** shall be fulfilled.

23 INTERNAL WIRING

This clause of Part 1 is applicable except as follows.

23.101 The insulation and sheath of internal wiring for the supply of magnetic valves and similar components incorporated in external hoses for connection to the water mains shall be at least equivalent to the electrical characteristics of light polyvinyl chloride sheathed flexible cord (code designation 60227 IEC 52).

Compliance is checked by the appropriate tests.

24 COMPONENTS

This clause of Part 1 is applicable except as follows.

24.1.4 Addition:

The number of cycles of operation for programmers is 3 000.

For lid or door interlocks, the number of cycles of operation declared for **6.10** and **6.11** of IEC 60730-2-12 shall not be less than 6 000. For washing machines that include a drying function, the minimum number of cycles of operation is increased to 9 000. If the interlock operates more than once during normal operation, the minimum number of cycles of operation is increased accordingly.

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24.101 Thermal cut-outs incorporated in washing machines for compliance with **19.4** shall not be self-resetting thermal cut-outs.

Compliance is checked by inspection.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CORDS

This clause of Part 1 is applicable.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of Part 1 is applicable.

27 PROVISION FOR EARTHING

This clause of Part 1 is applicable.

28 SCREWS AND CONNECTIONS

This clause of Part 1 is applicable.

29 CLEARANCES, CREEPAGE DISTANCES AND SOLID INSULATION

This clause of Part 1 is applicable except as follows.

29.2 Addition:

The microenvironment is pollution degree 3, and the insulation shall have a CTI not less than 250, unless the insulation is enclosed or located so that it is unlikely to be exposed to pollution during normal use of the appliance due to:

- a) condensation produced by the appliance; and
- b) chemicals, such as detergent or fabric conditioner.

30 RESISTANCE TO HEAT AND FIRE

This clause of Part 1 is applicable except as follows.

30.2 Addition:

For washing machines incorporating a programmer

or a timer, **30.2.3** is applicable. For other washing machines, **30.2.2** is applicable.

31 RESISTANCE TO RUSTING

This clause of Part 1 is applicable.

32 RADIATION, TOXICITY AND SIMILAR HAZARDS

This clause of Part 1 is applicable.

101 TESTS

101.0 Categories of Tests

Tests are classified as type, acceptance and routine tests.

101.1 Type Test

The tests specified in Table 102 shall constitute the type tests and shall be carried out on two samples of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the vacuum cleaners and water suction cleaning appliances shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction mechanical hazards, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of screw s and connections. The external surface finish shall be even and free from finishing defects.

101.1.1 Criteria of Acceptance

Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

Table 102 Schedule of Type Tests

(Clause 101.1)

SI No.	Test	Clause Reference
(1)	(2)	(3)
i)	Protection against access to live parts	8
ii)	Power input and current	10
iii)	Heating	11
iv)	Leakage current and electric strength at operating temperature	13
v)	Transient over voltages	14
vi)	Moisture resistance	15
vii)	Leakage current and electric strength	16
viii)	Overload protection of transformers and associated circuits	17
ix)	Abnormal operation	19
x)	Stability and mechanical hazards	20
xi)	Mechanical strength	21
xii)	Construction	22
xiii)	Internal wiring	23
xiv)	Components	24
xv)	Supply connection and external flexible cords	25
xvi)	Terminals for external conductors	26
xvii)	Provision for earthing	27
xviii)	Screws and connections	28
xix)	Clearances, creepage distances and solid insulation	29
xx)	Resistance to heat and fire	30
xxi)	Resistance to rusting	31

101.2 Acceptance Tests

The following shall constitute the acceptance tests:

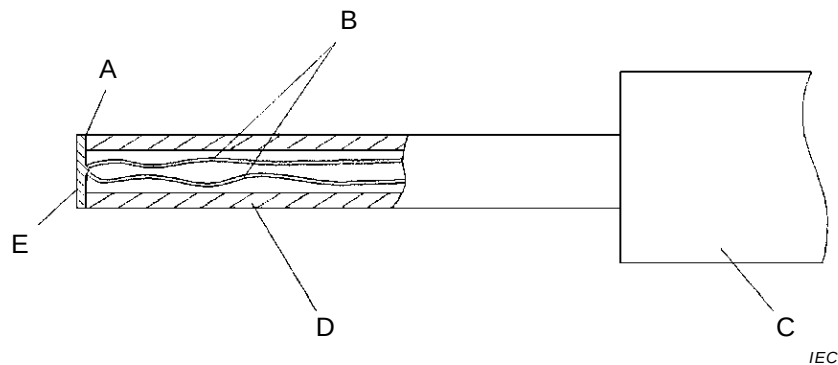
SI No.	Test	Clause Reference
(1)	(2)	(3)
i)	Protection against access to live parts	08
ii)	Power input and current	10
iii)	Heating	11
iv)	Leakage current and electric strength at operating temperature	13
v)	Moisture resistance	15
vi)	Leakage current and electric strength	16
vii)	Provision for earthing	27

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 h while conducting the test for moisture resistance (see 15).

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101.3 Routine Tests

Annex A of Part 1 shall be applicable for this clause.



Key

- A adhesive
- B thermocouple wires 0.3 mm diameter to IEC 60584-1 Type K (chrome alumel)
- C handle arrangement permitting a contact force of $4\text{ N} \pm 1\text{ N}$
- D polycarbonate tube: inside diameter 3 mm, outside diameter 5 mm
- E tinned copper disc: 5 mm diameter, 0.5 mm thick with flat contact face

FIG. 101 PROBE FOR MEASURING SURFACE TEMPERATURES

ANNEXES

The annexes of Part 1 are applicable except as follows.

ANNEX R

(normative)

SOFTWARE EVALUATION

R-2.2.5 Modification:

For programmable electronic circuits with functions requiring software incorporating measures to control the fault/error conditions specified in Table R.1, detection of a fault/error shall occur before compliance with **19**, **20.104**, **20.105**, **22.101** and **22.108** is impaired.

R-2.2.9 Modification:

The software and safety-related hardware under its control shall be initialized and shall terminate before compliance with **19**, **20.104**, **20.105**, **22.101** and **22.108** is impaired.

ANNEX AA

(Normative)

DETERGENT

The detergent specified in the instructions may be used, but if there is any doubt with regards to the test results, the composition of the detergent shall be as follows:

<i>Substance</i>	<i>Parts by mass percent</i>
Linear sodium alkyl benzene sulphonate (mean length of alkane chain C _{11.5})	6.4
Ethoxylated tallow alcohol (14 EO)	2.3
Sodium soap (chain length C _{12 to 16} : 13 % to 26 % and C _{18 to 22} : 74 % to 87 %)	2.8
Sodium tripolyphosphate	35.0
Sodium silicate (SiO ₂ : 76.75 % and Na ₂ O: 23.25 %)	6.0
Magnesium silicate	1.5
Carboxy methyl cellulose	1.0
Ethylenediamine tetra-acetic-sodium-salt	0.2
Optical whitener for cotton (dimorpholinostilbene type)	0.2
Sodium sulphate (as accompanying substance or added)	16.8
Water	7.8
Sodium perborate tetrahydrate (supplied separately)	20.0

NOTE — The composition of the detergent is extracted from IEC 60456.

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ANNEX BB

(normative)

AGEING TEST FOR ELASTOMERIC PARTS

The ageing test on elastomeric parts is carried out by measuring their hardness and mass before and after immersion in solutions of detergent and rinsing agent at elevated temperature.

The test is carried out on at least three samples of each part. The test procedure is as specified in IS 3400 (Part 6), with the following modifications.

BB-5 TEST LIQUIDS

Two test liquids are used:

- a) one liquid is obtained by dissolving 5 g of the detergent specified in Annex AA per litre of distilled water; and
- b) the other liquid is composed of 0.6 ml of rinsing agent as specified in 15.2 per litre of distilled water.

Care is to be taken to ensure that the total mass of the test pieces immersed does not exceed 100 g for each litre of solution, that the test pieces are completely immersed and that their entire surface is freely exposed to the solution. During the tests, the test pieces are not to be exposed to direct light. Test pieces of different compounds are not to be immersed at the same time in the same solution.

BB-6 TEST PIECES

BB-6.4 Conditioning

The temperature is $(23 \pm 2)^\circ\text{C}$ and the relative humidity is (50 ± 5) percent.

BB-7 IMMERSION IN THE TEST LIQUID

BB-7.1 Temperature

The solution is heated within 1 h with the test pieces

immersed, to a temperature of $75^{+5}_0^\circ\text{C}$ and maintained at this value. The solution is renewed every 24 h and heated in the same way.

NOTE — To avoid undue evaporation of the solution, a closed-circuit system or similar method can be used for renewing the solution.

BB-7.2 Duration

The test pieces are immersed for a total period of 48^{+1}_0 h.

The test pieces are then immediately immersed in a fresh solution which is maintained at ambient temperature. The pieces are immersed for (45 ± 15) min.

After having been removed from the solution, the test pieces are rinsed in cold water at $(15 \pm 5)^\circ\text{C}$ and then dried with blotting paper.

BB-8 PROCEDURE

BB-8.2 Change in Mass

The increase in mass of the test pieces shall not exceed 10 percent of the value determined before immersion.

BB-8.6 Change in Hardness

The micro-test for hardness applies.

The hardness of the test pieces shall not have changed by more than 8 IRHD. Their surface shall not have become sticky and shall show no crack visible to the naked eye or any other deterioration.

ANNEX CC
(normative)

DETERGENT FREE ELECTROLYSER WASHING MACHINES

The following modifications to this standard are applicable to washing machines for household and similar use that incorporate an electrolytic process employing an electrolyte instead of detergent.

NOTE — Additional subclauses and notes in this annex are numbered starting with 201.

CC-2 NORMATIVE REFERENCES

Addition:

<i>IS/ ISO/IEC Number</i>	<i>Title</i>
IS/IEC 60079-15	Explosive atmospheres: Part 15 Equipment protection by type of protection "n"
IEC 60068-2-52	Environmental testing — Part 2: Tests — Test Kb: salt mist, cyclic (sodium, chloride solution)

CC-3 TERMS AND DEFINITIONS

C-3.1.9 Addition:

Appliances employing an electrolyte added by the user shall be filled with the amount and type of electrolyte specified in the instructions.

CC-7 MARKING AND INSTRUCTIONS

CC-7.12 Addition:

The instructions for appliances intended to be filled with electrolyte by the user shall contain details of the electrolyte to be used and the substance of the following:

In order to avoid hazards, use only the electrolyte specified.

NOTE — Details of the electrolyte to be used can be given, for example, in terms of a generic name or a manufacturer's part number.

CC-7.12.1 Addition:

The instructions shall state that the appliance shall be installed so that there is a distance of at least 200 mm between the appliance enclosure and external heat sources, such as appliances containing heating elements.

CC-15 MOISTURE RESISTANCE

CC-15.2 Replace the Third and Fourth Paragraphs of the Compliance in the Replacement by the Following:

Appliances are operated under the conditions of **11**

but without a clothes load. When the maximum water level is reached, the inlet valve is held open and the filling is continued for 15 min after first evidence of overflow or until the inflow is automatically stopped by other means.

CC-15.101 Not Applicable.

CC-19 ABNORMAL OPERATION

CC-19.201 Appliances shall be constructed so that foaming does not affect electrical insulation.

Compliance is checked by the following test that is carried out immediately after **15.2**.

Detergent having a composition as specified in Annex AA is added, the quantity of detergent being twice the quantity of electrolyte necessary for normal washing. The appliance is then operated under the conditions specified in **11** but for one complete cycle with the programme that results in the longest period of operation.

The appliance shall then withstand the electric strength test of **16.3**.

CC-22 CONSTRUCTION

CC-22.6 Modification:

Instead of coloured water, a solution composed of 5 g of the detergent specified in Annex AA per litre of distilled water is used.

Add the following subclause:

CC-22.17 Addition:

Spacers intended to prevent the electrolyser aperture being blocked by walls shall be fixed so that it is not possible to remove them from the outside of the appliance by hand or by means of a screwdriver or a spanner.

CC-22.201 Appliances fitted with an electrolyser, consisting of cathodic and anodic chambers separated by an electrolytic separator, shall be constructed so that the electrolyser is always open to the atmosphere through an aperture of at least 5 mm in diameter, or 20 mm² in area with a width of at least 3 mm. The aperture shall be located so that it is unlikely to be obstructed in normal use.

Compliance is checked by inspection and by measurement.

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CC-22.202 During normal use of the appliance, the chemical reaction in the electrolyser shall not produce hydrogen gas that is released in hazardous amounts into areas

- a) where electrical components that produce arcs and sparks during normal operation or abnormal operation are mounted, unless these components have been tested and found at least to comply with IEC 60079-15 for group IIC gases; or
- b) that contain surfaces with a temperature exceeding 460 °C during normal operation or abnormal operation and that may be exposed to the released hydrogen gas.

Compliance is checked by inspection, by measuring the temperature of the relevant surfaces during normal operation and abnormal operation and by the following test.

The appliance is operated for one cycle under conditions of normal operation.

The concentration of hydrogen gas in the relevant areas is measured continuously for one wash cycle from the beginning of the test until the end of the cycle. The background hydrogen concentration measured prior to the test is subtracted from the maximum concentration measured during the test.

Instruments used for monitoring gas concentration, such as those that use infrared sensing techniques, shall have a fast response, typically 2 s to 3 s and shall not unduly influence the result of the test.

If gas chromatography is used, the gas sampling in confined areas shall occur at a rate not exceeding 2 ml every 30 s.

The measured value shall not exceed 50 percent of the lower flammability limit (LFL) of hydrogen.

NOTES

- 1 The LFL of hydrogen gas is 4 percent V/V of air.
- 2 Other instruments are not precluded from being used provided that they do not unduly influence the results.

CC-22.203 During normal use of the appliance, the chemical reaction in the electrolyser shall not produce wash water that causes corrosion due to the pH value of the wash water.

Compliance is checked by the salt mist test of IEC 60068-2-52, severity 2 being applicable. The pH value of the solution used shall be approximately equal to that of the wash water as measured during normal use of the appliance.

Before the test, enclosures having a coating are

scratched by means of a hardened steel pin, the end of which has the form of a cone with a top angle of 40°. Its tip is rounded with a radius of (0.25 ± 0.02) mm. The pin is loaded so that the force exerted along its axis is (10 ± 0.5) N. The pin is held at an angle of 80° to 85° to the horizontal and scratches are made by drawing the pin along the surface of the coating at a speed of approximately 20 mm/s. Five scratches are made at least 5 mm apart and at least 5 mm from the edges.

After the test, the appliance shall not have deteriorated to such an extent that compliance with this standard, in particular with 8 and 27, is impaired. The coating shall not be broken and shall not have loosened from the surface.

CC-29 CLEARANCES, CREEPAGE DISTANCES AND SOLID INSULATION

CC-29.2 Modification:

In the second dashed item of the addition, replace “detergent” by “electrolyte”.

CC-32 RADIATION, TOXICITY AND SIMILAR HAZARDS

Addition:

The ozone concentration produced by the chemical reactions in the electrolyser shall not be excessive.

Compliance is checked by the following test, which is carried out in a room without openings having dimensions of 2.5 m × 3.5 m × 3.0 m, the walls being covered with polyethylene sheet.

The room is maintained at approximately 25 °C and 50 percent relative humidity. The appliance is positioned in accordance with the instructions and then operated for one cycle under conditions of normal operation.

The ozone sampling tube is to be located 10 mm from the gas outlet aperture specified in **CC.22.201**. The background ozone concentration measured prior to the test is subtracted from the maximum concentration measured during the test.

The percentage of ozone in the room shall not exceed 5×10^{-6} .

Annex BB

Modification:

Instead of the solution containing detergent, a solution of the electrolysed portion of the wash water obtained under the conditions of **11** is used.

ANNEX DD
(informative)

WASHING MACHINES INCORPORATING A POWER DRIVEN WRINGER

The following modifications to this standard are applicable to washing machines for household and similar use that incorporate a power driven wringer.

NOTE — Additional subclauses and notes in this annex are numbered starting with 201.

DD-7 MARKING AND INSTRUCTIONS

DD-7.1 Addition:

The safety release mechanism of power-driven wringers shall be marked to indicate its method of operation, unless its operating means has to be continuously actuated by the user.

DD-7.12 Addition:

The instructions for washing machines incorporating a power-driven wringer shall draw attention to the potential hazards involved when operating the wringer and shall state that

- a) the wringer must be disengaged or switched off when not in use; and
- b) the appliance must not be operated by children.

DD-11 HEATING

DD-11.7 Addition:

The appliance is operated for three cycles, with a rest period of 4 min between cycles. Each cycle consists of washing followed by wringing.

The duration of each wringing period is 8 min. The wringer is loaded by passing a board through the rollers once a minute, the roller pressure being adjusted to the maximum value. The board is approximately 20 mm thick and 800 mm long, its width being at least equal to three-quarters of the effective length of the rollers. The board is uniformly tapered at each end down to a thickness of approximately 3 mm, over a distance of 200 mm.

DD-19 Abnormal Operation

DD-19.7 Addition:

Moving parts of a wringer are locked even if a trip bar prevents rotation of the rollers.

DD-20 STABILITY AND MECHANICAL HAZARDS

DD-20.201 Power-driven wringers shall be constructed so that the pressure between the rollers has to be maintained by the user, unless a readily accessible safety release or other means of protection is incorporated.

The release mechanism shall operate easily without violent ejection of any part and shall release pressure on the rollers immediately. The rollers shall separate either by at least 45 mm at both ends or by at least 25 mm at one end and 75 mm at the other.

The safety release shall be operable by a person standing in any normal working position relative to the wringer, even if the fingers of both hands are trapped between the rollers.

Power-driven wringers shall be constructed to prevent fingers being squeezed between a roller and the frame.

Power-driven wringers shall be controlled by an easily accessible switch.

NOTE — The switch controlling the washing machine can also control the wringer.

Compliance is checked by inspection, by measurement, by manual test and by the following test.

The pressure between the rollers is adjusted to its maximum value. The board described in **11.7** is passed between the rollers and the wringer is stopped when the board is approximately halfway through. A force is gradually applied to the operating means of the safety release. The release shall operate before the force exceeds 70 N.

(Continued from second cover)

If an appliance within the scope of this standard also incorporates functions that are covered by another Part 2 of IS 302, the relevant Part 2 is applied to each function separately, as far as is reasonable. If applicable, the influence of one function on the other is taken into account.

This standard is a product family standard dealing with the safety of appliances and takes precedence over horizontal and generic standards covering the same subject.

NOTE — Horizontal and generic standards covering a hazard are not applicable since they have been taken into consideration when developing the general and particular requirements for the IS 302 series of standards.

An appliance that complies with the text of this standard will not necessarily be considered to comply with the safety principles of the standard if, when examined and tested, it is found to have other features which impair the level of safety covered by these requirements.

An appliance employing materials or having forms of construction differing from those detailed in the requirements of this standard may be examined and tested according to the intent of the requirements and, if found to be substantially equivalent, may be considered to comply with the standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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Eastern : 8 th Floor, Plot No 7/7 & 7/8, CP Block, Sector V, Salt Lake, Kolkata, West Bengal 700091	{ 2367 0012 2320 9474
Northern : Plot No. 4-A, Sector 27-B, Madhya Marg, Chandigarh 160019	{ 265 9930
Southern : C.I.T. Campus, IV Cross Road, Taramani, Chennai 600113	{ 2254 1442 2254 1216
Western : Plot No. E-9, Road No.-8, MIDC, Andheri (East), Mumbai 400093	{ 2821 8093

Branches : AHMEDABAD. BENGALURU. BHOPAL. BHUBANESHWAR. CHANDIGARH. CHENNAI. COIMBATORE. DEHRADUN. DELHI. FARIDABAD. GHAZIABAD. GUWAHATI. HIMACHAL PRADESH. HUBLI. HYDERABAD. JAIPUR. JAMMU & KASHMIR. JAMSHEDPUR. KOCHI. KOLKATA. LUCKNOW. MADURAI. MUMBAI. NAGPUR. NOIDA. PANIPAT. PATNA. PUNE. RAIPUR. RAJKOT. SURAT. VISAKHAPATNAM.